UNIVERSITY OF LAY ADVENTISTS OF KIGALI

UNILAK RWAMAGANA CAMPUS

IDFinder Rwanda

Lost and Found Identity Document Recovery System

Milestone 2

Requirements Engineering and System Analysis

Group 2

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1 Introduction

This report specifies the proposed requirements for IDFinder Rwanda, a system intended to connect reports of found national identity cards and driving licences with people searching for their original documents. Milestone 1 identified recovery difficulty through informal discussions and used three illustrative scenarios to examine possible problems in searching, reporting and handover. Milestone 2 translates those concerns into a defined scope, numbered functional requirements, measurable quality criteria and user stories with acceptance conditions.

The specification follows the lecturer's requirements engineering approach in slides 27–36 and Section 6 of the Student Project Guide. It explains the users, operating assumptions, proposed behaviour and conditions under which that behaviour will be considered acceptable. It also includes the original use-case diagram and two detailed use-case descriptions. The aim is to give a development team enough information to understand the intended pilot and connect each feature to a particular discovery concern.

The requirements remain proposed. No Milestone 2 user validation sessions were documented in the supplied material. The report therefore provides a requirements baseline for review without presenting planned user sessions as completed research. Its expanded explanations clarify the original presentation, while the numbered functional requirements and their priorities retain the original pilot direction.

1.1 Purpose and Intended Users

The main purpose of IDFinder Rwanda is to support the recovery of an original document through a traceable report, search, claim and handover process. A finder needs to make the discovery known through an appropriate channel. A document owner needs to identify a possible match without exposing private information. Staff at a participating collection point need to review reports, check a claimant's ownership in person and record the return of a document.

1.2 Definitions Used in the Specification

A found report is a record describing a document discovery. A listed report is a found report approved for public search. A claim is a private request by a signed-in owner who believes a listed document may belong to them. A reference is the identifier used to distinguish a report. A collection point is the participating location responsible for holding the physical document

and performing the handover process. A handover is the staff record confirming that a document has been collected after the claim was approved.

An approved listing indicates that staff have accepted the report for publication. It does not itself establish that a claimant owns the document. An approved claim follows an in-person ownership check. Keeping these two decisions separate helps prevent confusion between making a found report searchable and deciding who can collect the document. A closed listing indicates that the documented handover has been completed.

2 System Scope

2.1 Included Activities

The pilot covers national identity cards and driving licences. It includes recording a found report, reviewing the report before publication, searching approved listings and submitting a private claim for a possible match. It also includes displaying the claim status to the authenticated claimant, checking ownership in person and recording one handover against an approved claim. Finder access to the status of their own report is included as a Should-have feature.

The public search will use document type, approximate area and date range. Public results will expose only the type, approximate area, date and report reference. These details support discovery of a possible match while keeping the claim and ownership check separate. A search result will therefore provide an entry into the private recovery process rather than publicly confirming the identity of the holder.

2.2 Activities Outside the Pilot

The release excludes issuing replacement documents, integration with government identification registries and automatic decisions about ownership. Public display of document images, complete identity numbers and personal contact details is also excluded. The original presentation identifies these boundaries so that the pilot remains focused on recovery of the physical original and verification through a participating collection point.

Optional status notifications are treated as a Could-have enhancement. They will depend on an opt-in choice and will contain no sensitive details. The core reporting, claim and return journey must remain usable without this enhancement. The pilot scope also assumes that

physical verification remains a staff responsibility, with the system recording the decision and outcome.

3 Overall Description and Operating Assumptions

3.1 User Roles

The finder submits information about a found document and receives a reference. Where the finder status feature is implemented, the signed-in finder can see whether their report is awaiting review, listed or closed. The document owner searches public listings and signs in before submitting or viewing a private claim. The owner can follow the current claim status but cannot make an ownership decision or record collection.

Collection-point staff review reports assigned to their location. They approve or reject a report, consider claims through an in-person ownership check and record a handover after approval. Their authority is restricted to their assigned collection point. Staff at a different location and unrelated owners must not receive private claim details. These boundaries make the role and record relationship part of the access decision.

3.2 Operating Environment and Dependencies

The pilot will be accessed through a web interface and will depend on a working internet connection. The presentation does not prescribe a framework, hosting provider or database product. The requirements therefore describe behaviour independently of those implementation choices. Public search and authenticated claim handling must present a consistent view of the report's current state.

A participating collection point must be able to hold documents and check claimants. This is an operating assumption rather than a confirmed partnership. Before operational deployment, the team must establish who is responsible for receiving documents, what ownership checks are accepted and how staff accounts are assigned. A digital report must correspond to a document that the receiving location can locate and return.

3.3 General Process Rules

A new report first awaits staff review. Approval makes its limited listing searchable, while rejection prevents public publication. A claimant must sign in and select an approved report before submitting a private claim. The claim remains available to its owner and the staff

responsible for the collection point. In-person checking precedes the approval or rejection of ownership, and an approved claim is required before a handover can be recorded.

The return record must be created once for the document. If a repeated request attempts to record the same handover again, the system must reject it and preserve the original record. After a successful return, the related listing and claim close so that the document is no longer presented as available for collection. These rules connect the integrity criterion to an observable change in the recovery journey.

4 Functional Requirements

Each functional requirement states what the system shall do and is identified by an FR code. The Must requirements describe the complete reporting-to-return journey. FR-08 is a Should requirement because it improves finder follow-up after the core workflow has been implemented. The explanations below describe how the original requirement supports its intended user.

4.1 FR-01 Found Report Submission

The system shall accept a found report with document type, area and date, and issue a reference.

The finder will provide the details needed to distinguish the discovery. Required fields must be validated before the report is saved. A successful submission will create a report awaiting staff review and show its reference. The reference will allow the report to be identified consistently during review and later enquiries. This Must requirement addresses the need for a structured reporting route.

4.2 FR-02 Search of Approved Reports

The system shall filter approved reports by document type, area and date range.

The search will return only approved listings matching the selected filters. Pending and rejected reports must remain outside the public results. An empty result must be presented clearly so that the owner can understand that no listed match meets the chosen criteria. A result is a possible match requiring further checking, and the public fields remain limited by NFR-01. This is a Must requirement.

4.3 FR-03 Private Claim Submission

The system shall let an owner submit a private claim for an approved report.

The owner must be authenticated before making a claim. The claim will be linked to the selected report and routed to the responsible collection-point staff. Claim details will not be included in public listings. Staff approval of the report will not automatically approve the claim, because ownership must be checked separately. This Must requirement supports a controlled transition from searching to verification.

4.4 FR-04 Claim Status Visibility

The system shall display the current claim status to its authenticated owner.

The owner will be able to open their claim and see its current position in the review process. The same access route must not expose another person's claim. The status must reflect the stored staff decision and eventual closure after collection. This Must requirement addresses uncertainty about follow-up after a claim is submitted.

4.5 FR-05 Staff Review of Found Reports

The system shall let assigned staff review, approve or reject found reports.

Only staff assigned to the relevant collection point may make the publication decision. Approval changes the report from awaiting review to listed, making its restricted public details searchable. Rejection keeps it outside the public listing. This Must requirement supports screening before publication and separates the finder's submission from staff acceptance.

4.6 FR-06 Ownership Decision

The system shall let assigned staff approve or reject a claim after checking ownership in person.

The in-person check is a precondition for the decision. Staff will record whether the claim is approved or rejected, and the owner will see the resulting status through FR-04. An unsuccessful ownership check must not lead to release of the document. This Must requirement defines the human decision on which the later handover depends.

4.7 FR-07 Single Handover Record

The system shall let assigned staff record one handover against an approved claim.

The claim must already be approved and the document must still be available for collection. Successful confirmation closes the claim and listing. A second request must preserve the first record and produce no additional return entry. This Must requirement supports a reliable outcome record and prevents repeated collection confirmations.

4.8 FR-08 Finder Report Tracking

The system shall let a finder view whether their report is awaiting review, listed or closed.

The finder must use their authenticated account and report reference to view their own status. The displayed state will allow the finder to understand whether the report has been accepted and whether the document has been returned. This Should requirement supports follow-up without granting the finder access to a claimant's private verification details.

5 Non Functional Requirements

The quality criteria specify how the proposed behaviour will be evaluated. They are acceptance targets for the pilot rather than measured achievements. Each criterion has an NFR identifier and an associated verification method, allowing the later testing milestone to assess the system against a stated condition.

5.1 NFR-01 Privacy

Public results expose only type, approximate area, date and reference. Inspect the UI and response payload.

Verification will examine both what the page displays and what the application returns to the browser. Removing private fields from the visible page alone is insufficient if the response still contains them. The public search test must therefore confirm that document images, complete identity numbers, contact details and claim information are absent from public results.

5.2 NFR-02 Access Control

Unassigned staff and unrelated owners receive no claim details in negative access tests.

The test will use an authorised owner, a different owner, assigned staff and staff from another collection point. It will attempt to read the same claim through the interface and direct requests. OWASP advises validating permission on every request (OWASP Foundation, n.d.). This supports checking access at the point of data retrieval as well as during navigation.

5.3 NFR-03 Performance

At least 95% of searches return within 2 seconds in 50 trials on a test set of 1,000 reports.

The test will record elapsed response times in a defined test environment and report its network and server conditions. At least 48 of the 50 searches must complete within the threshold. The test data should contain different document types and report states so that search filtering can be exercised. This target does not represent a claim about actual production performance.

5.4 NFR-04 Accessibility

The core forms have visible labels, keyboard access and text contrast of at least 4.5:1.

The reporting, searching and claim forms will be checked for readable labels and keyboard operation. The review will also measure the contrast of normal text and check whether focus can be followed when navigating without a mouse. The criterion draws on the keyboard and minimum contrast provisions in WCAG 2.2 (World Wide Web Consortium, 2024). It addresses selected criteria rather than claiming complete WCAG conformance.

5.5 NFR-05 Integrity

A second attempt to record handover is rejected and the original record stays intact.

Testing will confirm a handover, repeat the confirmation and compare the stored records. A concurrent confirmation attempt must also leave a single successful return. The original confirmation and the closed state must remain consistent. A rejected duplicate must not reopen the listing or create another collection record.

6 User Stories and Acceptance Criteria

The eight user stories express the functional requirements from the perspective of the person performing the task. Each acceptance condition states a starting situation, the action taken

and the observable result. The original story IDs are retained so that the backlog, use-case descriptions and traceability section refer to the same behaviour.

6.1 US-01 Reporting a Found Document

Priority Must. As a finder, I want to report a found document so that the discovery can enter the recovery process.

Given valid report details, when I submit them, then I receive a reference.

If required details are missing, the system identifies the affected fields and saves no report. This story is linked to FR-01 and test T01.

6.2 US-02 Finding a Possible Match

Priority Must. As an owner, I want to find possible matches so that I can identify a document that may be mine.

Given a listed match, when I search using the relevant filters, then I see a masked result.

Pending and rejected reports must be excluded, and private details must remain hidden. This story is linked to FR-02 and test T02.

6.3 US-03 Making a Private Claim

Priority Must. As an owner, I want to claim a match privately so that the responsible staff can check ownership.

Given a listed report, when I submit a claim while signed in, then staff receive the claim.

An unauthenticated submission must not create a claim. A report that is no longer available must not be offered for a new collection request. This story is linked to FR-03 and test T03.

6.4 US-04 Checking Claim Progress

Priority Must. As an owner, I want to check progress so that I can understand the current status of my claim.

Given my claim, when I open it, then I see its current status.

The same request from an unrelated account must disclose no claim details. This story is linked to FR-04 and test T04.

6.5 US-05 Screening a Report

Priority Must. As assigned staff, I want to screen reports so that only accepted reports become publicly searchable.

Given a pending report, when I approve it, then it becomes searchable.

Rejecting the report must keep it outside the public results. Staff from an unassigned point must be unable to approve it. This story is linked to FR-05 and test T05.

6.6 US-06 Deciding a Claim

Priority Must. As assigned staff, I want to review ownership so that a collection decision follows an in-person check.

Given an in-person ownership check, when I record my decision, then the claim status updates.

If ownership cannot be verified, the claim is rejected and the document remains in custody. This story is linked to FR-06 and test T06.

6.7 US-07 Recording Return

Priority Must. As assigned staff, I want to record return so that the completed recovery has a clear outcome.

Given an approved claim, when I confirm pickup, then the case closes.

A second confirmation must be rejected and leave the original handover intact. This story is linked to FR-07 and test T07.

6.8 US-08 Following a Found Report

Priority Should. As a finder, I want to track my report so that I can see whether it is awaiting review, listed or closed.

Given my reference and account, when I open my report, then I see its status.

The finder must not receive private claim details through this status view. This story is linked to FR-08 and test T08.

7 MoSCoW Prioritisation and Backlog

The Must category contains US-01 to US-07 because the pilot needs the full route from found reporting to verified return. Omitting any of these steps would leave a gap in publication, search, private claiming, ownership checking or collection. These stories must be scheduled together as the core release. The original presentation does not assign story points, so the team must estimate the work when transferring the stories to the live backlog.

The Should category contains US-08, which lets a finder follow the status of their report after submission. It improves the experience but follows the reporting-to-return path in implementation priority. Optional notifications are a Could feature, subject to opt-in and limited message content. Replacement issuance, registry integration and automatic verification remain in the Won't category for this release.

The implementation sequence follows the dependencies between stories. Report submission and staff screening establish the records that public search can return. Private claims and owner status follow that approved listing. The staff decision and single handover complete the journey. The sequence helps organise development while preserving the need to demonstrate the complete Must workflow within the course build period.

8 Use Case Diagram

The use-case diagram shows the finder, document owner and collection-point staff outside the IDFinder Rwanda boundary. The finder submits a found report and, where the Should feature is included, tracks its status. The owner searches reports, submits a private claim and views claim status. Staff review reports, decide claims and record handovers. The solid association lines indicate participation in those activities.

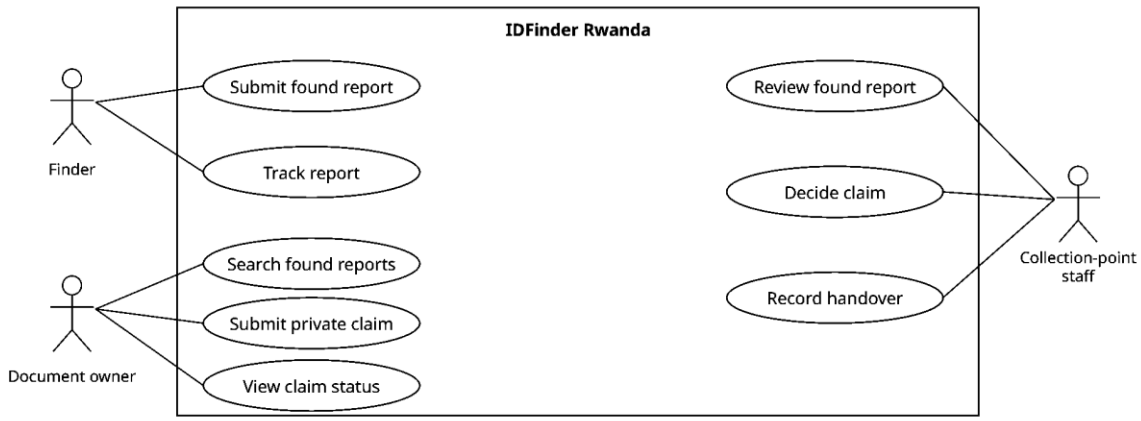


Figure 1 IDFinder Rwanda use-case diagram reproduced from the presentation in black and white

The diagram summarises the user goals, while the following descriptions explain their preconditions and alternative outcomes. A sign-in requirement appears in the use-case conditions rather than as a separate goal in the original figure. Ownership decisions remain with assigned staff. The diagram does not imply that the application can verify ownership through an external registry or release a document without a physical check.

9 Detailed Use Case Descriptions

9.1 UC-01 Submit a Found Report

The primary actor is the finder. The use case supports FR-01 and connects to staff review under FR-05. Its precondition is that the finder has signed in and has basic information about the found document. The trigger is the finder’s decision to record the discovery through the reporting form.

In the main flow, the finder chooses the document type and enters the area and date. The system validates the required information, saves a report awaiting review and returns its reference. The finder can use that reference to identify the submission. Assigned collection-point staff then review the report, and approval makes the restricted listing available to public search.

In the alternative flow, required information is missing or invalid. The system identifies the affected fields and saves no report. The finder may correct the details and submit again. If staff reject a saved report during review, it remains outside the searchable listings. The

successful submission outcome is a pending record, while searchable publication is a later result of the separate staff decision.

9.2 UC-02 Claim and Collect

The primary actor is the document owner, supported by collection-point staff. The use case links FR-03, FR-04, FR-06 and FR-07. The owner must be signed in and must have identified a possible match in an approved report. The collection point must be able to locate the physical document and perform its ownership checking process.

In the main flow, the owner selects the report and submits a private claim. Staff receive the claim and conduct the in-person ownership check. If the check is satisfactory, staff record approval. At collection, assigned staff confirm the handover once. The system closes the claim and listing, allowing the owner to see the completed status and preventing another recorded handover for the same return.

In the alternative flow, ownership cannot be verified. Staff reject the claim and retain the document in custody. The owner can see the decision through the authenticated status view. If a repeated handover request arrives after collection, the system rejects it and preserves the original record. If another successful return has already closed the report before collection, the system must not record an additional handover.

10 Requirements Traceability

The original Milestone 2 presentation groups requirements under five design concerns. These groupings reorganise the Milestone 1 themes around the proposed workflow. Reporting includes the finder’s route into the system. Search addresses uncertainty about a possible match. Private claims address controlled ownership checking, return addresses completion and follow-up addresses visibility of the current outcome.

10.1 D1 Reporting

D1 is linked to FR-01 and FR-05, which correspond to US-01 and US-05. T01 verifies that a valid submission produces a reference and that missing details prevent saving. T05 verifies that staff approval publishes the restricted listing while rejection keeps it outside search. Together these tests cover the transition from discovery reporting to an accepted public record.

10.2 D2 Search

D2 is linked to FR-02, US-02 and T02. The test begins with an approved report matching the selected filters. It checks that the result appears with only the permitted public details and its reference. It also checks that pending and rejected reports are excluded. This link connects the owner's need to know where to check with a specific observable search result.

10.3 D3 Private Claims

D3 is linked to FR-03 and FR-06, corresponding to US-03 and US-06. T03 verifies authenticated private claim submission. T06 verifies the assigned staff decision after in-person checking and confirms that an unsuccessful check leaves the document in custody. Access tests under NFR-02 accompany both activities because claim privacy depends on the requesting account and collection-point assignment.

10.4 D4 Return

D4 is linked to FR-07, US-07 and T07. The test confirms that an approved claim can receive one handover record and that this record closes the claim and listing. A repeated or concurrent confirmation must not produce another return. The link connects the need for a reliable outcome to the integrity criterion in NFR-05.

10.5 D5 Follow Up

D5 is linked to FR-04 and FR-08, corresponding to US-04 and US-08. T04 checks the owner's access to the current claim status. T08 checks the finder's access to the status of their own found report when that Should feature is implemented. These tests address the uncertainty that can remain after an initial report or claim has been submitted.

11 User Validation and Submission Evidence

The first review journey is intended for a document owner. The participant will examine the Must requirements, a masked search result and the private claim steps. The discussion should establish what information helps them identify a possible match and whether the status information explains the next action. Any requested change must be recorded against the affected requirement and story so that the team can explain how feedback changes the specification.

The second review journey is intended for a finder or a collection-point worker. The review will examine the reporting fields, custody assumptions, ownership decision and handover record. It must clarify who can verify a claimant and how staff will know that a document is available for collection. The two reviews serve different purposes and should preserve those differences when findings are documented.

No completed sessions or participant responses were provided for this milestone. The user review requirement therefore remains pending. Likewise, the source presentation contains story specifications but no live Jira or Trello board link or screenshot. These items must be added as submission evidence after the activities take place. The final repository submission should place the documentation in the milestone folder and identify the submitted state through the m2 release tag, as required by the lecturer.

12 Conclusion

The proposed requirements define a limited document recovery journey through reporting, screening, search, private claiming, ownership checking and recorded handover. The eight functional requirements and five quality criteria establish the behaviour and acceptance targets for the pilot. The user stories and traceability explanations connect that behaviour to the concerns explored in Milestone 1, while the use-case diagram and descriptions show who performs each task.

The next approval decision depends on whether the participating collection arrangement is workable and whether the Must features address the needs of the users who review them. The specification provides a detailed baseline for that review. Actual user feedback, backlog evidence and confirmed operating responsibilities will complete the evidence needed to assess readiness for the design milestone.

References

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